HW

Find out the list of companies that are for placement drive using DSA.

Amazon, Microsoft, Google, Adobe, Samsung, Intel, Walmart Global Tech, Oracle, NVIDIA, Paypal, American express

Find out Gate syllabus which is related to DSA (for CSE)

Unit 1: Programming and Data Structures

Introduction to Data Structures

Arrays, Linked lists, Stacks, Queues, Hash Tables, Trees, Graphs, Recursion

Unit 2: Algorithms

Asymptotic Analysis, Searching Algorithms, Sorting Algorithms, Hashing, Divide and Conquer, Greedy Algorithms, Dynamic Programming, Graph Algorithms

Find out set of Inputs and outputs for the following

i) Electricity Bill

Inputs:

Consumer Number
Consumer Name
Billing Month
Previous Meter Reading
Current Meter Reading
Tariff Type

Outputs:

Units consumed
Energy charges
Fixed charges
Taxes (if any)
Total Bill Amount
Due Date
Bill status

ii) Bus Ticket

Inputs:

Passenger Name
Source
Destination
Travel Date
Number of passengers

Bus type
Age
Gender
o/p:
Boarding point
Booking status
Fare
Ticket Number
Bus Number
Arrival Time
Seat Numbers(s)
Departure Time

iii, Message to a Friend

Inputs:
Sender Name
Friend's Name/contact
Message Text
Date and Time

o/ps:
Message Displayed
Sent status (sent/Delivered)
Delivery Time
Read status (seen / unread)

4) Today when you started from home to college, what are the various data structures you identified in your surroundings

Data structures	Real-Life Example
Array	Row of packed bikes and cars
Queue	line of people at bus stop
Stack	stack of books
Graph	Road connecting different places
linked list	Bus route map
Hash table	Dictionary / phone contacts
Tree	Branches of tree or college Hierachy
Matrix (2D array)	classroom seating arrangement
priority Queue	Ambulence in traffic
Circular Queue	Vehicles moving around a traffic roondabout.